Neuromorphic Vertical Stability Control via Petri-Net-to-Spiking Compilation: A Reduced-Order Evaluation

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Abstract

We describe a software pipeline that maps continuous-marking stochastic Petri nets to leaky integrate-and-fire populations and a bounded actuator interface. The implementation makes the symbolic policy, compilation products and runtime state inspectable, but its current physical evidence is deliberately narrow. We evaluate the implemented PID, observer-based LQR and push–pull spiking controllers on the same deterministic reduced-order vertical-instability model. For a growth rate of $200\,\mathrm{s}^{-1}$, a 5 mm kick and a 1 kHz control update, PID and LQR settle in 25 ms and 39 ms. The real spiking controller does not stabilise the case and reaches approximately 160 mm peak displacement. Its internal dynamics lag the 5 ms instability e-folding time. This is a material negative result: the current SNN is not competitive for this fast-VDE case.

We preserve the exact output of an earlier workstation-latency harness and explain why it is historical rather than ranking evidence. Its tracked JSON contains five lane aggregates, but not the exact invocation episode count or seed; it also used different plant/action paths and accepted noise and delay arguments without injecting them. A subsequent schema-v3 one-episode run demonstrates common scenario, full-coil action and policy-only timing wiring, but is explicitly promotion-ineligible: every completed lane crosses the disruption threshold at time zero and NMPC-JAX is unavailable. We therefore pre-register, but do not execute, a fresh 200-episode real-kernel cohort with fail-closed admission gates. No FPGA, neuromorphic-device or radiation experiment has been performed.

Keywords: Petri net; spiking neural network; tokamak vertical stability; neuromorphic control; stochastic computing; negative result

1 Motivation and boundary

Elongated tokamak plasmas can be unstable to the axisymmetric vertical mode, requiring active feedback on timescales set by the resistive structure and the closed-loop plant [Lazarus et al., 1990, Humphreys et al., 2015]. Feedback stabilisation of tokamak plasma instabilities is established practice on devices around the world [Walker et al., 2006]. Neuromorphic implementations are interesting only if their timing, robustness and verification properties are measured under a physically meaningful contract.

This paper is bound to commit `8df066b60bf7`; the full revision is in the paper metadata. The plant is a declared reduced-order ordinary differential equation. The work is not a discharge experiment, hardware-in-the-loop validation, radiation test or production PCS. No comparison with an independently tuned facility controller is attempted.

2 Stochastic Petri-net representation

A continuous-marking Petri net is represented as

$$\mathcal{N} = (P, T, W_{\text{in}}, W_{\text{out}}, \mathbf{m}_0), \tag{1}$$

where P and T are finite sets of places and transitions, W_{in} and W_{out} are non-negative arc-weight matrices, and $\mathbf{m}_0 \in [0,1]^{|P|}$ is the initial marking. A transition is enabled when its weighted inputs satisfy a declared threshold. Firing consumes and produces marking according to the two arc matrices, followed by the runtime's explicit bounded-domain policy.

The construction is inspired by classical Petri-net semantics [Petri, 1962, Murata, 1989], while the continuous $[0,1]$ marking is chosen to align with stochastic-number representations [Gaines, 1969, Alaghi and Hayes, 2013]. Figure 1 shows the eight-place policy used as a software fixture. It is a control-policy graph, not a complete plasma model.

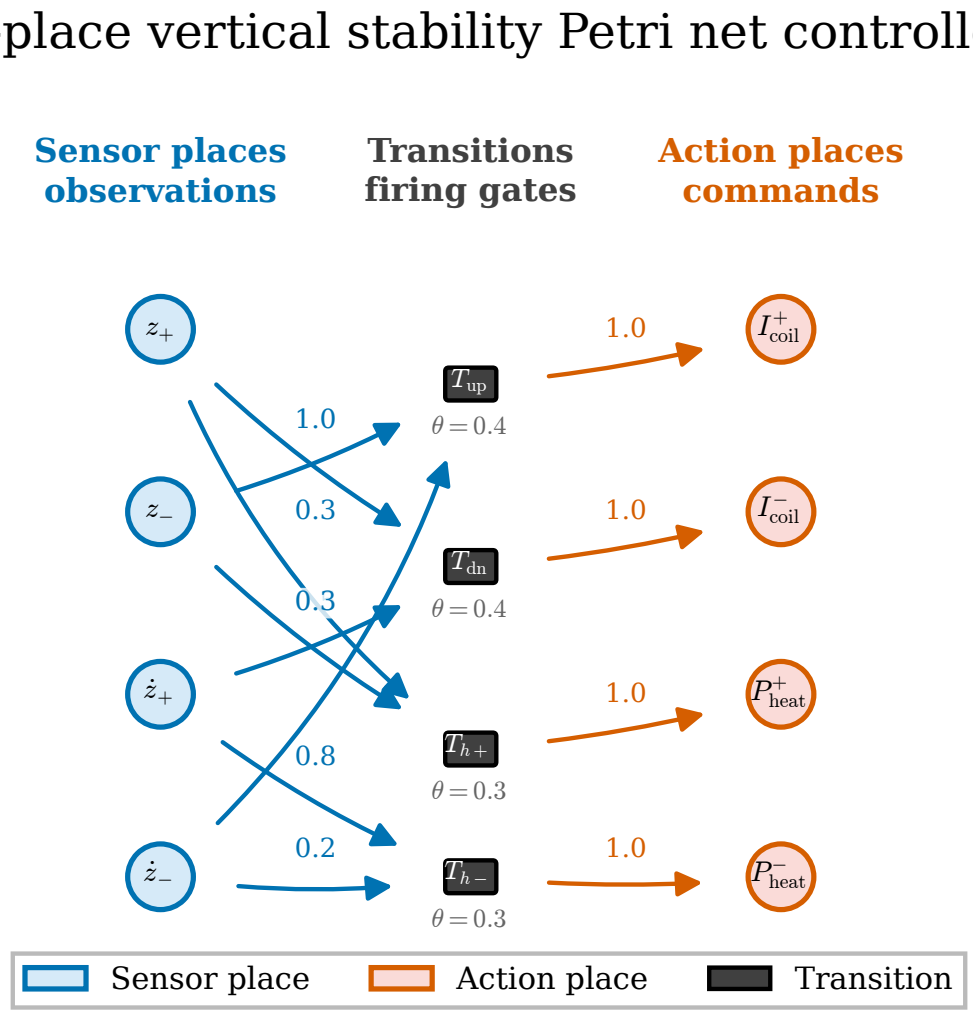


Figure 1: Eight-place, four-transition Petri-net policy used by the compiler fixture. The diagram describes software state flow; it is not a facility safety case.

3 Compilation and runtime

The pipeline has three observable stages:

1. a builder validates places, transitions, arcs, thresholds and initial marking;
2. the compiler maps places and transitions to population and connection structures, including stochastic bitstream representations;
3. the runtime injects observations, advances symbolic/spiking state and decodes an action through the actuator boundary.

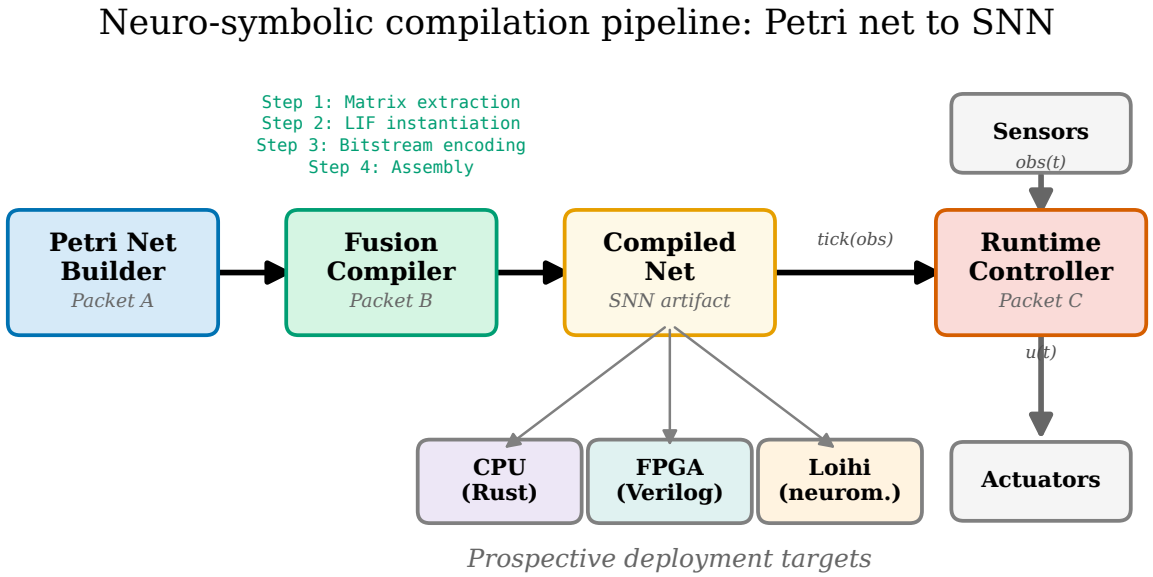


Figure 2: Software compilation path from policy graph to runtime artefact. Hardware targets are prospective; this draft reports software execution.

The neuron model follows a leaky integrate-and-fire form [Maass, 1997],

$$\tau_m \frac{dV}{dt} = -(V - V_{\text{rest}}) + RI(t), \tag{2}$$

with threshold, reset and refractory behavior. Figure 3 is a pedagogical deterministic trace; it is not a hardware waveform.

The public runtime validates finite inputs and clips decoded actions at the actuator boundary. Software tests exercise builder validation, compilation, runtime behavior and safety interlocks. They do not prove equivalence to an unbuilt FPGA or neuromorphic circuit.

4 Reduced-order vertical benchmark

4.1 Plant and protocol

The benchmark uses the declared linearised vertical plant

$$\ddot{z} = \gamma^2 z - \frac{k_{\text{coil}}}{m_p} u, \tag{3}$$

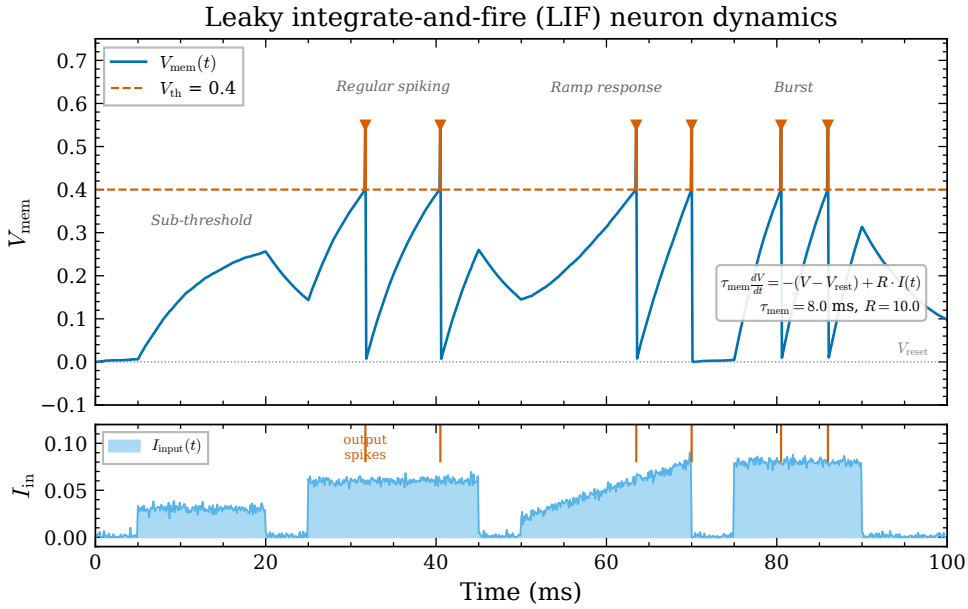


Figure 3: Pedagogical LIF response generated by the paper-local script.

with $\gamma = 200\,\mathrm{s}^{-1}$, $k_{\mathrm{coil}} = 0.5$, $m_p = 1$, a 1 ms integration/control step and a ± 10 kA command limit. A 5 mm displacement is injected at 10 ms and the simulation runs for 60 ms. The 5 ms open-loop e-folding time makes this a deliberately fast case.

The PID gains are chosen from a stated second-order target. The LQR lane uses the repository's observer-based state-space controller. The SNN lane uses the actual 64-neuron push–pull `SpikingControllerPool`; it is not a relabelled PID trace. All three trajectories are produced by `figures/fig_vertical_stability.py`.

4.2 Result

Table 1: Deterministic reduced-order response. "Settling" is the first time after the last excursion outside the ± 0.1 mm band.

Controller	Peak $\lvert z \rvert$ (mm)	Settling (ms)	Outcome
PID, retuned	5.00	25	stabilises this model
LQR with observer	5.20	39	stabilises this model
Push–pull SNN	≈ 160	—	diverges

The negative result is not a tuning footnote. With the present dynamics, the spiking path reacts too slowly for the 5 ms instability time. Faster synapses, a different coding scheme or a hybrid architecture may change the result, but those are new hypotheses requiring new predeclared experiments.

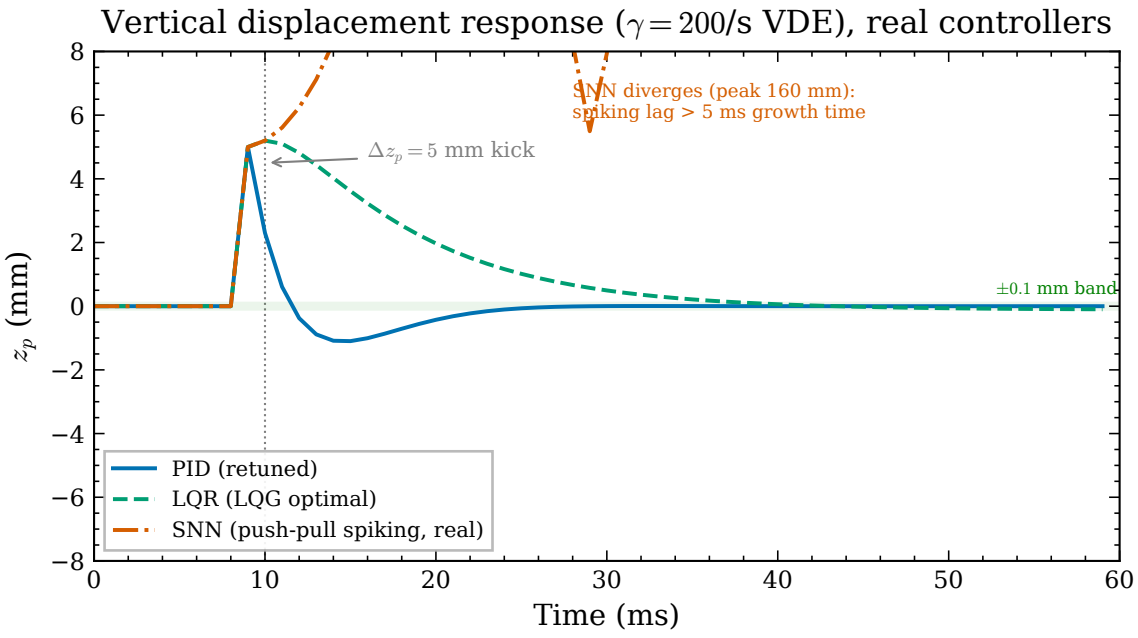


Figure 4: Same-plant response of the implemented controllers. The SNN curve leaves the displayed range and peaks near 160 mm. This is a reduced-order numerical experiment, not a tokamak discharge.

5 Evaluation-contract evolution

5.1 Historical methodology and preserved output

The legacy result was first tracked at commit `1163ce454b75` and remains recoverable as Git blob `5616f192a7b4`; the exact source JSON has SHA-256 `394be6c32a56...`. Table 2 reproduces all five serialized p50/p95 values. This preservation does not rehabilitate the old comparison.

At implementation revision `476908debdd8`, Python PID and H-infinity used the Python equilibrium flight simulator, Rust-PID used a separate Rust reduced-order simulator, NMPC-JAX used an untrained random dynamics model and returned only its first scalar action, and Nengo-SNN received the scalar error as radial input with zero vertical input. Latency covered a complete lane-specific `run_shot` divided by reported steps, not one policy call. The harness accepted and printed its noise and delay arguments but did not pass them into episode runners. It serialized neither a seed, a common disturbance trace, exact invocation episode count, failure ledger nor environment custody. The code default was 1000 episodes, but the exact invocation count that produced the tracked aggregate is unknown and is reported as unknown.

5.2 Why the contract changed

The historical data remain useful for auditability and for showing what the software emitted at that time. They cannot support a speedup, accuracy, disruption, energy-efficiency or deployment claim because the lanes were not accuracy-equivalent. The replacement contract consequently requires one plant, one R/Z observation, one full ordered coil-current-offset action in MA, the same actuator delay/lag/slew/saturation, one pre-materialized PCG64 disturbance trace per episode, exactly one timed policy call per control step, explicit failures, and digest-bound recovery.

Table 2: Historical legacy-harness latency output. The exact invocation episode count was not serialized; lanes are not accuracy-equivalent.

Legacy lane	p50 (μs)	p95 (μs)
PID	3431.376	3624.250
Rust-PID	0.523	0.670
H-infinity[†]	3227.145	3606.877
NMPC-JAX	45450.300	49773.092
Nengo-SNN	23573.109	24736.161

[†]Invalidated stale scalar-plant calibration. All other rows remain historical orientation only.

5.3 Schema-v3 transition run

The exact schema-v3 report at `8df066b60bf7` is a one-episode surrogate wiring assessment, not a reduced-cost substitute for real-plant evidence. Six lanes completed against the same disturbance digest; NMPC-JAX remained visibly unavailable because it lacks a calibrated dynamics artefact and held-out gate. Every completed lane had the same mean radial error (4.0387 m), mean vertical error (3.7877 m), disruption rate (1.0) and disruption time (0.0 s). The last quantity proves that the present initial state is already outside the declared 0.5 m criterion before actuation can discriminate between policies.

Table 3: One-episode schema-v3 wiring assessment. Policy-only timing is recorded, but the run is incomplete and promotion-ineligible.

Lane	Status	p50 (μs)	p95 (μs)
PID	complete	19.471	30.855
Rust-PID	complete	16.598	26.484
H-infinity	complete	80.413	107.266
LQR	complete	65.463	88.523
MPC	complete	682.805	802.083
LIF-NEF-SNN	complete	2332.483	2925.647
NMPC-JAX	unavailable	—	—

The policy-only timings in Table 3 verify instrumentation and lane wiring only. They cannot select a controller because the run has one episode, uses the non-promotable surrogate scope, starts disrupted and omits an admissible NMPC lane.

5.4 Pre-registered fresh cohort

The paper-local `fresh_cohort_protocol.json` freezes a 200-episode, seed-42, 30 s, real-FusionKernel comparison across PID, H-infinity, LQR, MPC, NMPC-JAX, LIF-NEF-SNN and Rust-PID. The primary endpoints are disruption rate and radial/vertical tracking error; tracking reward, actuator effort and policy-only p50/p95/p99 latency are secondary. Comparisons are paired by the per-episode disturbance-trace digest, failures remain failures, and no lane can be promoted from latency alone.

The cohort is prepared but intentionally blocked. Before execution, the real-kernel initial equilibrium must pass the predeclared time-zero criterion, NMPC-JAX must load a provenance-bound calibrated model with a disjoint held-out gate, and all seven lanes must be available. The primary run then requires a dedicated unloaded host and the frozen recovery identity; an independent

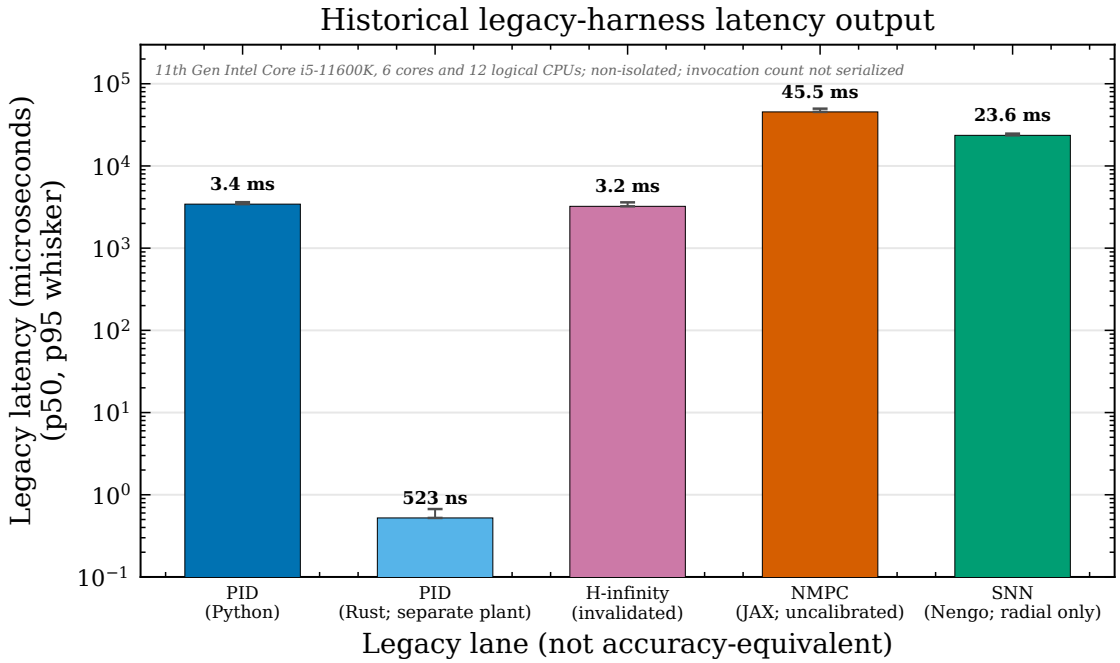


Figure 5: Historical legacy-harness output, p50 with a p95 whisker. The source values are preserved exactly, including the invalidated H-infinity lane. Different plant/action paths and an unknown invocation count prohibit a current controller ranking.

second host must reproduce scientific metrics while host-dependent timing is compared separately. No long-running fresh cohort is reported in this manuscript.

6 Stochastic encoding and radiation boundary

In a unipolar stochastic code, a value $x \in [0, 1]$ is represented by the fraction of ones in a bitstream of length L. Independent streams allow an AND gate to estimate multiplication. A single flipped bit changes the decoded value by at most $1/L$, an encoding-level sensitivity rather than a radiation qualification result.

No FPGA timing closure, power measurement, single-event-upset campaign or neuromorphic-device run is included. Earlier quantitative hardware latency and power estimates are therefore absent from the results.

7 Limitations and next experiments

The current evidence is bounded by the following facts:

- the fast-VDE SNN result is negative;
- the plant is a deterministic reduced-order model and lacks a full plasma, vessel, power-supply, sensor and communication stack;
- the historical latency output lacks a serialized exact invocation count, seed and common lane contract;

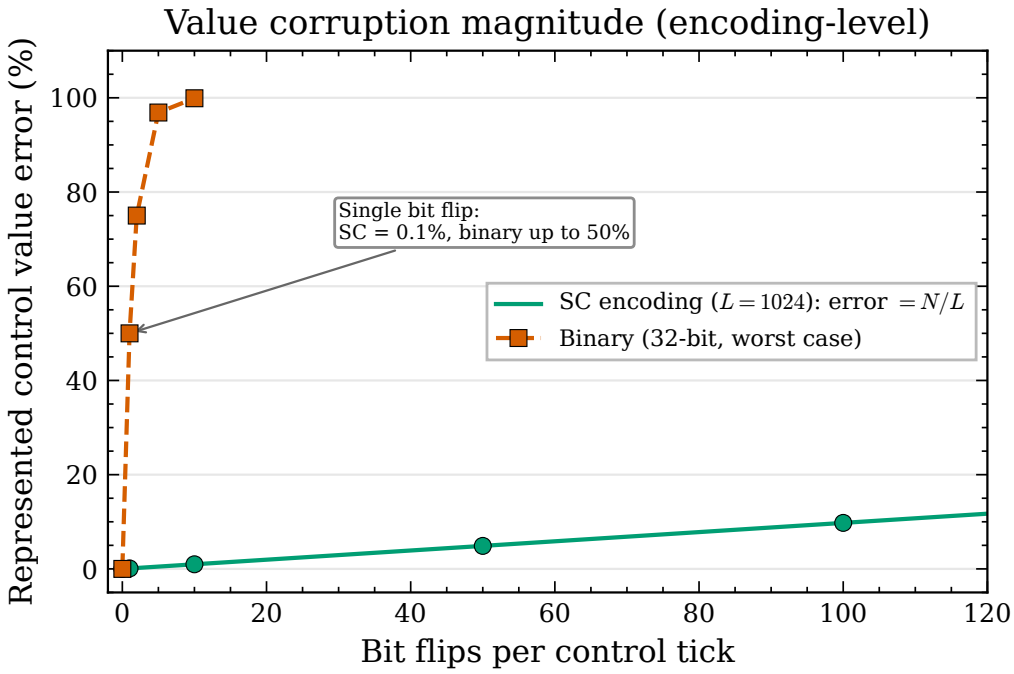


Figure 6: Analytical encoding-error illustration. The plotted arrays are not irradiation measurements and do not establish device-level fault tolerance.

- the schema-v3 transition run is a single wiring-only surrogate episode whose completed lanes are disrupted at time zero;
- NMPC-JAX has no calibrated dynamics artefact or held-out validation gate and is therefore unavailable;
- the pre-registered fresh cohort has not run and lacks independent second-host reproduction;
- symbolic/runtime tests are not a hardware refinement proof;
- radiation and dedicated-hardware properties are unmeasured;
- no experimental control, hardware-in-the-loop or production PCS claim is made.

The next credible sequence is to freeze an accuracy-equivalent reduced-order controller cohort, reproduce it independently, add sensor/actuator and delay uncertainty, and only then synthesize and measure a hardware candidate. A device experiment must report timing, power, numerical/control quality and fault injection together.

8 Reproducibility and availability

This directory contains the manuscript, bibliography, figure generators, generated figures, the Git-object-bound historical result, the exact schema-v3 wiring assessment, the blocked fresh-cohort protocol, generated LaTeX tables and build instructions. Code is available at https://github.com/anulum/scpn-fusion-core. The review draft is bound to commit `8df066b60bf7`; the full revision is recorded in the paper metadata. The manuscript-content licence has not yet been selected by the author; repository code retains its existing licence.

9 Conclusion

The Petri-net-to-spiking path is implemented and inspectable, but the current SNN does not stabilise the declared fast vertical-instability case. This result keeps the software architecture distinct from demonstrated control performance. The paper-local evidence preserves the legacy record without treating it as a current ranking, exposes the failed schema-v3 admission state, and pre-registers the next accuracy-equivalent study without claiming that it has run.

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